Scalable MoE-LoRA: Progress Report

Efficient Routing for Fine-Grained Expert Mixtures

Anh Trung Nguyen Jiaqi Lu Andrew Huang

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Motivation

Sparsely-activated Mixture-of-Experts (MoE) architectures are standard at frontier scale (DeepSeek-V3, Mixtral, Qwen-MoE). The *granularity* of an MoE layer — how a fixed parameter budget is split between many small experts or few large ones — is hypothesized to matter because with K experts and top-k activation, a token can address $\binom{K}{k}$ possible expert combinations. At K=8, k=2 that is 28 combinations; at K=64, k=16 it is $\approx 4.9 \times 10^{14}$. Fine granularity gives the model exponentially more ways to specialize *at the same active-parameter budget per token*.

A controlled study at pretraining scale would burn tens of thousands of GPU-hours, which we do not have. Instead, we run the study in the **LoRA (parameter-efficient fine-tuning)** setting on a frozen LLaMA 3.2 1B base: only the MoE adapter and its router train, so any quality difference is attributable to the adapter alone. Strong consistent signals trigger full-pretraining follow-up; weak signals save that compute. The granularity and router axes here are defined identically to what a full-MoE practitioner chooses, so the principles transfer.

Phase A — Exploratory LoRA and MoE-LoRA work (complete)

We began with a 4-variant reference table on a 12-dataset subset, seed 42: standard LoRA (Hu et al., 2021), MoE-LoRA (Luo et al., 2024, K independent expert pairs with linear routing), TM-LoRA (token-modulated LoRA: shared A, B with additive per-token expert-vector modulation), and our unified shared-bottleneck RoutedLoRA. Phase A validated the infrastructure (18-dataset + 7-OOD evaluator, SLURM harness on Adroit) and gave early signal that a shared-bottleneck rank-1 routed LoRA at K=64 outperformed both plain LoRA and loop-MoE-LoRA on reasoning (0.592 vs 0.574 vs 0.571). The four variants differed in K, r, top-k, architecture, and router simultaneously, so the signal was noisy — but it motivated the controlled factorial below.

Phase B — Granularity × router factorial, seed 42 (complete)

Design. Nine LLaMA 3.2 1B runs on the 18-dataset reasoning + NLG suite, 3 epochs, effective batch 36. All routed configs share the exact same LoRA matrix parameter count ($K \cdot r$=64) and active matrix FLOPs per token (top_k $\cdot\, r$=16); only the (K, r) split and the router mechanism vary. Two routers: standard linear ($\mathcal{O}(d \cdot K)$ per layer) and low-rank ($\mathcal{O}(d \cdot r_{\text{dim}} + K \cdot r_{\text{dim}})$).

Result. Fine granularity is monotonically beneficial in-distribution for both router types (Table **??**). The linear router substantially outperforms low-rank on OOD (by 2–3 pts), a larger gap than the 0.3–0.5 pt in-distribution gap — routing quality matters most under distribution shift. The best OOD performer is `r2_k32_linear` (0.255); interestingly on the linear side r2_k32 slightly edges out the finest r1_k64 on OOD despite losing in-distribution. Exception: the coarsest linear run (K=8, top_k=2) diverged in epoch 2 — likely routing collapse at very few active experts, flagged for a load-balancing re-run. OOD numbers are low in absolute terms because the 6 benchmarks include hard tests (BBH, MMLU-Pro, GPQA Diamond) where a 1B LoRA is close to random; the relative ordering is what matters.

Phase C (OOD). Each of the 9 Phase B checkpoints was also evaluated on 7 held-out OOD benchmarks (MMLU, MMLU-Pro, BBH, IFEval, AGIEval, GPQA Diamond, TruthfulQA); the right column of Table **??** is the mean of the 6 accuracy-metric subsets (IFEval is BLEU/ROUGE-L and is excluded from the mean). Fine granularity does not harm OOD — the routed configs all beat baseline OOD — but the *router* choice matters more under distribution shift than it does in-distribution.

Phase D — Router comparison at fine granularity (complete)

With granularity settled, the remaining bottleneck for scaling MoE is the router's own parameter cost: a full linear router at DeepSeek-V3 scale (d=7168, K=256) needs ∼1.8M params per layer. Phase D compares four additional routers at K=64, r=1, top_k=16, holding everything else constant (same LoRA matrices, data, seed, optimizer):

- **Product-key** (Lample et al. 2019): two scorers $R^d \to R^{\sqrt{K}}$; score for expert (i, j) is $s_1[i] + s_2[j]$; top-k over the full $\sqrt{K} \times \sqrt{K}$ product space. Cost $\mathcal{O}(d\sqrt{K})$. Any expert reachable.

Config	K	r	top_k	Router	val_loss	in-dist	OOD
baseline	—	8	—	standard LoRA	1.984	0.500	0.207
granularity_r8_k8	8	8	2	lowrank rdim=16	1.951	0.509	0.226
granularity_r4_k16	16	4	4	lowrank rdim=16	1.949	0.522	0.223
granularity_r2_k32	32	2	8	lowrank rdim=16	1.944	0.526	0.230
granularity_r1_k64	64	1	16	lowrank rdim=16	1.933	0.531	0.217
granularity_r4_k16_linear	16	4	4	linear	1.926	0.523	0.226
granularity_r2_k32_linear	**32**	**2**	**8**	**linear**	**1.926**	**0.531**	**0.255**
granularity_r1_k64_linear	**64**	**1**	**16**	**linear**	**1.923**	**0.534**	**0.245**
granularity_r8_k8_linear *	8	8	2	linear	2.139	0.371	0.197

Table 1: Phase B + C: granularity sweep on LLaMA 3.2 1B, seed 42. OOD is the mean accuracy across 6 held-out benchmarks (MMLU, MMLU-Pro, BBH, AGIEval, GPQA Diamond, TruthfulQA). *Training diverged in epoch 2 (routing collapse at top_k=2 with linear routing), flagged for re-run with load balancing.

- **Hierarchical**: group router plus within-group router shared across selected groups; two-level top-k. Same $\mathcal{O}(d\sqrt{K})$ cost but with a group bottleneck.
- **Cosine**: lowrank rdim=16 with L2 normalization on query and keys. Tests whether the linear-vs-lowrank gap is score-scale instability.
- **Early-shared**: a single linear router computes top-k from the first LoRA layer's input; all 32 subsequent injection points reuse the cached decision. Motivated by the Omni-Router Transformer finding.

Router	Per-layer cost (d=2048, K=64)	val_loss	in-dist	OOD
linear (Phase B reference)	131K	1.923	0.534	**0.245**
lowrank rdim=16 (Phase B ref.)	34K	1.933	0.531	0.217
product_key (Phase D)	33K, $\mathcal{O}(d\sqrt{K})$	1.924	**0.536**	0.224
cosine (Phase D)	34K	1.954	0.520	0.230
hierarchical (Phase D)	33K, $\mathcal{O}(d\sqrt{K})$	1.954	0.517	0.225
early_shared (Phase D)	131K/$L \approx$ 4K amortized	1.952	0.523	0.200

Table 2: Phase D: router comparison at K=64 on LLaMA 3.2 1B, seed 42. All rows share identical LoRA matrices; only the router varies. OOD is mean over 6 accuracy-metric benchmarks.

Findings. *Product-key routing matches full linear on val_loss and in-distribution at $\mathcal{O}(d\sqrt{K})$ parameter cost*: 0.001 val_loss gap, +0.002 in-distribution (product-key slightly ahead). On OOD, product-key (0.224) is noticeably better than the low-rank router at the same parameter-cost scale (0.217) but still ∼2 pts behind full linear (0.245). At K=64 that is 4× cheaper than linear; at K=4096, d=7168 (frontier-MoE territory) it extrapolates to 23×. Hierarchical's shared-within-group constraint is too restrictive. Cosine normalization alone does not close the linear-vs-lowrank gap on either metric. The linear router's OOD advantage is the most robust Phase D finding — *the in-distribution vs OOD split is the key observation: $\sqrt{K}$-cost routing is sufficient for the training distribution, but full-capacity routing pays off under distribution shift.* Early-shared ties on val_loss and in-dist but is the worst on OOD (0.200), suggesting per-layer routing functions as an implicit distribution-shift defense — a follow-up-worthy finding on its own.

Phase E + F — Multi-seed and 3B scale-up (running)

Two follow-ups are queued. **Phase E** re-runs baseline and product_key at seeds 0 and 123, giving mean $\pm\sigma$ over 3 seeds — the bridge from "clear single-seed trend" to "statistically defensible claim" that both reviewers and a full-pretraining commit decision need. **Phase F** scales baseline and product_key to LLaMA 3.2 3B at seed 42 with gradient checkpointing. The 3B VRAM stress test passed at 69–70% peak (under the 75% cap); training is auto-gated on that and will fire. Expected wall ∼3–4 days.

Open questions for discussion

- **Full MoE pretraining follow-up.** If Phase E confirms the Phase D ranking with error bars and Phase F shows the findings persist at 3B, the next step is to integrate product-key routing into a full (non-PEFT) MoE pretraining run. Sensible given remaining compute? What model / dataset scale is most informative?
- **Early-shared routing deserves its own study.** A single decision shared across 32 layers landing within

0.01 val_loss of per-layer routing challenges a common MoE assumption. Worth a standalone paper on routing-decision frequency?

- **Routing collapse at K=8, top_k=2 linear.** Known load-balancing issue or something specific to rank-8 experts with hard top-2? Re-run with auxiliary balancing loss planned.
- **Cheap analyses on existing data.** Per-dataset routing-profile clustering (is fine routing learning a task taxonomy?) and generalization-gap shape vs granularity. Both strengthen the paper.
- **Other router designs.** Mixture-of-routers, learned-hashing, ReXMoE-style cross-layer expert sharing — worth fitting into the same framework before the paper deadline?

Repository: https://github.com/attrung/scalable-moe-lora. Full phase-by-phase plan, reproducibility instructions, and raw result JSONs are on the `main` *branch.*